

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S5 (S, FE) / S5 (PT) (S) Examination June 2024 (2019 Scheme)

Course Code: CST 301**Course Name: FORMAL LANGUAGES AND AUTOMATA THEORY**

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

- | | | |
|----|--|---|
| 1 | Draw transition diagram for NFA (with or without ϵ -moves) for strings starting with '10' or '11'. $\Sigma = \{0,1\}$ | 3 |
| 2 | Define the language acceptability by DFA, NFA and ϵ -NFA | 3 |
| 3 | Write down the regular expression for strings with even number of a's.
$\Sigma = \{a, b\}$ | 3 |
| 4 | How can we identify two equivalent states in a DFA? | 3 |
| 5 | Is the grammar ' $E \rightarrow E+E / E * E / a$ ' ambiguous? Justify your answer. | 3 |
| 6 | State Myhill - Nerode Theorem | 3 |
| 7 | Differentiate between DPDA and NPDA | 3 |
| 8 | List out the transition rules ' δ ' used to convert given CFG G to its equivalent NPDA M . | 3 |
| 9 | Explain how Turing Machine differs from PDA. | 3 |
| 10 | Why TM model is widely accepted and considered as a general model for computers? | 3 |

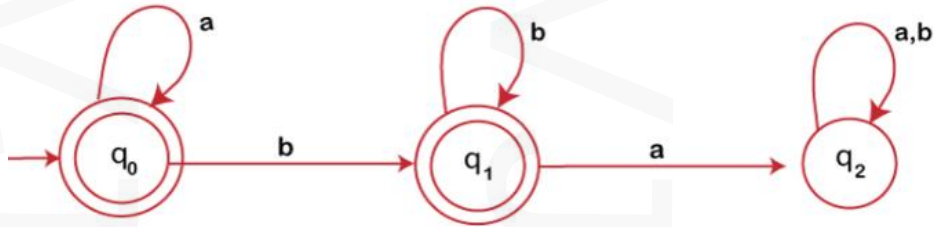
PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

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|----|--|---|
| 11 | a) Prove that for every NFA M , there exist an equivalent DFA M' | 7 |
| | b) Design an NFA (without ϵ -moves) for strings ending with '01'. Convert it into equivalent DFA. $\Sigma = \{0, 1\}$ | 7 |
| 12 | a) Draw the transition diagram of DFA for $L = \{ baxb / x \in \{a, b\}^* \}$ and obtain its regular grammar | 7 |

- b) Prove that, if L is accepted by an ordinary NFA, there exist an equivalent ϵ -NFA that also accepts L

Module -2

- 13 a) Obtain the regular expression equivalent to the following DFA



- b) Develop equivalent automata for the Regular Expression $(ab+ab)^*(a+bb)^*a^*$
- 14 a) Prove that for every Regular Expression ' R ', there is an ϵ -NFA ' M '
- b) Using pumping lemma, show that $L = \{a^p / p \text{ is a prime number}\}$ is not regular

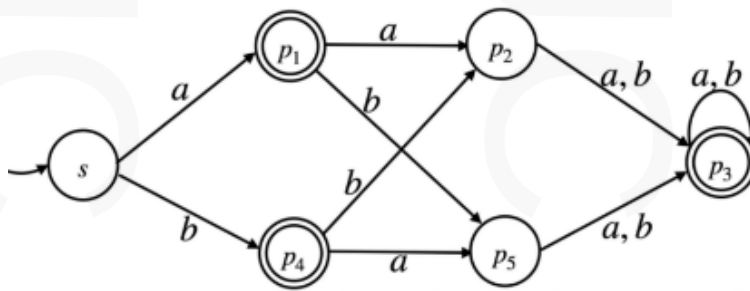
Module -3

- 15 a) What is Greibach Normal Form (GNF)? Convert the following CFG to GNF

$$S \rightarrow ABA, \quad A \rightarrow aA / \epsilon, \quad B \rightarrow bB / \epsilon$$

- b) a) Design CFG for the following languages
- $L = \{wcw^R / w \in \{a, b\}^*\}$ where w^R represents reverse of w
 - Strings with exactly 2 zeros. $\Sigma = \{0, 1\}$
 - $(011+1)^*(01)^*$

- 16 a) Minimize the following DFA using Myhill – Nerode theorem



- b) Show the equivalence classes of Canonical Myhill – Nerode relation for the languages of binary strings with odd number of zeros and even number of ones.

Module -4

- 17 a) Prove that for every PDA accepted by empty stack, there exists an equivalent PDA accepted by final state.
- b) Design PDA for $L = \{a^i b^j c^k / k = i + j\}$. Illustrate the working for the string 'aabbccccc'

- 18 a) Design PDA for $L = \{ ww^R / w \in \{a, b\}^* \}$. 7
b) State and prove any three closure properties of Context Free Languages. 7

Module -5

- 19 a) Design TM for $L = \{ a^n b^n / n > 0 \}$. Trace out the working for the input string 'aabb' 7
b) Explain Chomsky hierarchy for formal languages and evaluate various types 7
- 20 a) Design a TM to copy a block of zeros to the right side, leaving one blank symbol (b) in between. Assume that initially the input tape contains $b0^n b$ and TM halts with $b0^n b0^n b$ as the tape content. 7
b) Explain the working of Universal Turing Machine with its encoding scheme 7
